FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
CONCEPT RECAPITULATION TEST – I
PAPER –2
TEST DATE: 26-04-2023

ANSWERS, HINTS & SOLUTIONS

Physics

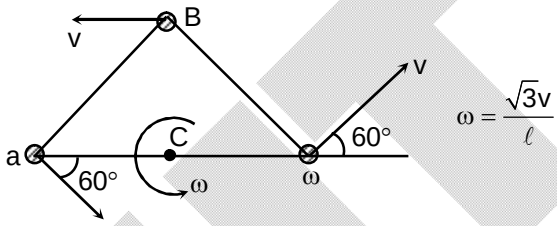
PART – I

Section – A

1. D

Sol. Displacement of G = Displacement w.r.t. C + Displacement of C

Further, by mathematical calculations, it can be solved.

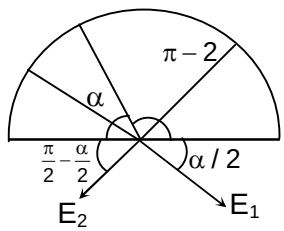


2. C

Sol. $E_1\cos\frac{\alpha}{2} = E_2\sin\frac{\alpha}{2}$ …(i)

$E_1\sin\frac{\alpha}{2} + E_2\cos\frac{\alpha}{2} = E_0$ …(ii)

Solving we get E_1



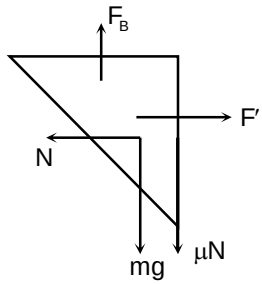
3. A

Sol. For equilibrium

N = F′

and, $F_B = \mu N + mg$

where, $F' = \left(\frac{h}{2}\right)(\rho g)(hb)$ and $F_B = \frac{1}{2}(h)(h\tan\theta)b\rho g$

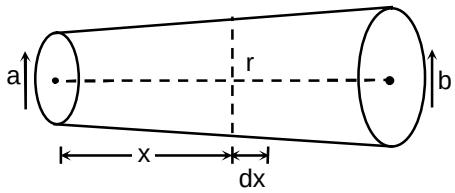


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4. D

Sol. $r = a + \left(\frac{b-a}{\ell}\right)x$

$\Delta\ell = \int_0^{\ell}\left(\frac{F}{\pi r^2}\right)\frac{dx}{Y}$

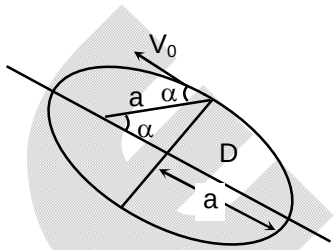


5. A, B, C

Sol. Total energy $E = \frac{1}{2}mv^2 - \frac{qQ}{4\pi\varepsilon_0 a} = \frac{-qQ}{8\pi\varepsilon_0 a}$

$E = -\frac{qQ}{4\pi\varepsilon_0(2a)}$

$\therefore$ The particle is moving on an ellipse of semi-major axis a and the time period will $T = 2\pi\sqrt{\frac{a^3}{\left(\frac{qQ}{4\pi\varepsilon_0 m}\right)}} = \sqrt{\frac{16\pi^3 ma^3}{qQ}}$



From geometry $b = a\sin\alpha$

$\therefore$ Area A = $\pi ab = \pi a^2 \sin\alpha = \frac{\sqrt{3}}{2}\pi a^2$

6. A, C

Sol. Path difference = 0

$\frac{d^2}{D} = \frac{yd}{2D} - \left(\frac{\mu_2}{\mu_3} - 1\right)t$

y = 2 mm

When slab is removed then path difference = $\frac{d^2}{D} - \frac{y_1 d}{2D} = 0$, $y_1 = 2d = 4mm$

7. A, B, D

Sol. $U = \frac{-2GMm}{\sqrt{d^2+y^2}} + \frac{-GM^2}{2d}$

$= \frac{-2GMm}{\sqrt{d^2+y^2}} - \frac{2Gm^2}{d}$

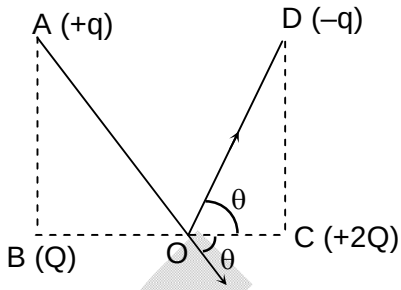
$= -2Gm^2\left[\frac{2}{\sqrt{d^2+y^2}} + \frac{1}{d}\right]$

$= |\Delta U| = \frac{1}{2}mv^2$

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8. C, D

Sol. For ball O (positively charged) to be in equilibrium at the middle point of BC



$$\left[\frac{q}{4\pi\in_0 (a\sqrt{5})^2}+\frac{q}{4\pi\in_0 (a\sqrt{5})^2}\right]\cos\theta+\frac{Q}{4\pi\in_0 a^2}=\frac{2Q}{4\pi\in_0 a^2}$$

$$\Rightarrow \quad q=\frac{5\sqrt{5}Q}{2}\left[\text{as}\cos\theta=\frac{a}{a\sqrt{5}}=\frac{1}{\sqrt{5}}\right]$$

If $q=\frac{5\sqrt{5}Q}{2}$ ball will be in equilibrium, even if charges are inter charged.

If charge at C is +Q, then

$$\frac{2q\cos\theta}{5}+Q=Q$$

$$\frac{2q\cos\theta}{5}=O$$

$a \neq 0$ $\theta = 90°$ which is not possible

9. A, B, C

Sol. Rate of loss of heat = K $(T - T_0)$

$$12=K\times 30\Rightarrow k=\frac{2}{5}$$

$$ms\frac{dT}{dt}=12-\frac{2}{5}(T-15°)$$

$$\Rightarrow \int_{15}^{25}\frac{dT}{90-2T}=\frac{1}{(2.5)s}\int_0^{300}dt \qquad \Rightarrow \ s=\frac{240}{\ln\left(\frac{3}{2}\right)}$$

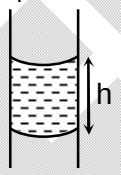
10. A, C

Sol. As the two prisms are identical and their refracting angle are inverted the net dispersion and net deviation of the ray by the combined prisms will be zero.

Section – B

11. 4

Sol.



$$P_0-\frac{2T}{R}+\rho gh-P_0=\frac{2T}{R}$$

$$h=\frac{4T}{R\rho g}$$

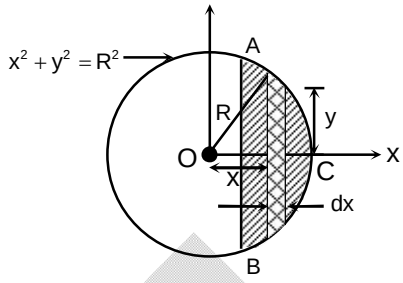
12. 9

Sol. As we know that electrostatic field at any point P, due to uniformly distributed line change (if point does not lie on the line) is along the angle bisector of the angle made by line charge at point P, so $\theta = 45° \Rightarrow$ a = 4 and b = 5 $\Rightarrow$ a + b = 9.

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13. 5

Sol. For the moment of inertia of the about disc of mass dm at distance x about the given axis, we will use



$$d\ell = \frac{1}{2}dmy^2, dm = \rho dV$$

Where, y will be radius

Density of spherical segment is

$$\rho = \frac{M}{V} = \frac{M}{\int_{\frac{R}{2}}^{R}\pi y^2 dx} = \frac{M}{\int_{R/2}^{R}\pi(R^2 - x^2)dx}$$

$$\Rightarrow \quad \rho = \frac{M}{\frac{5\pi R^3}{24}} = \frac{24M}{5\pi R^3}$$

$$d\ell = \frac{1}{2}dm \cdot y^2 d\ell$$

$$d\ell = \frac{1}{2}\rho\pi y^2 dx \times y^2 = \frac{\rho\pi}{2}y^2 dx$$

Total moment of inertia of segment of sphere is

$$\ell = \int d\ell = \frac{\rho\pi}{2}\int_{\frac{R}{2}}^{R} y^2 dx, \qquad \ell = \frac{\rho\pi}{2}\int_{\frac{R}{2}}^{R}(R^2 - x^2)^2 dx$$

$$= \frac{\rho\pi}{2}\int_{\frac{R}{2}}^{R}(R^4 - 2R^2x^2 + x^4)dx$$

$$\ell = \frac{24M}{5\pi R^3} \times \frac{\pi}{2} \times \left[R^4 x - \frac{2R^2x^3}{3} + \frac{x^5}{5}\right]_{R/2}^{R}$$

$$= \frac{24M}{5\pi R^3} \times \frac{\pi}{2} \times \frac{53R^5}{480} \ell = \frac{53}{200}MR^2$$

$$\ell = \frac{53}{200}MR^2 = \frac{53mR^2}{40x}$$

$$\Rightarrow \quad x = 5$$

14. 9

Sol. $g_h = \dfrac{gR^2}{(R+h)^2}$. Given $g_h = \dfrac{g}{100}$, $\dfrac{gR^2}{(R+h)^2} = \dfrac{g}{100}$ or $R + h = 10$ R or $h = 9R$.

15. 8

Sol. The stresses are zero at the free end and maximum at the axis. Therefore, the rod will ruptures at the middle.

Let us consider the element of rod at a distance x from the axis, the mass of element

$dm = \rho A dx$

Applying Newton's second law,

$F - (F + dF) = (\rho A dx)\omega^2 x$

Where ω is the rotation speed.

$$F = -\int dF = -\rho A\omega^2 \frac{x^2}{2} + C$$

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Rupture of rod will occur when $f = \sigma$

$$\therefore \frac{\rho\omega^2 l^2}{8} = \sigma \quad \Rightarrow \omega = \sqrt{\frac{8\sigma}{\rho l^2}} \Rightarrow n = \frac{1}{2\pi}\sqrt{\frac{8\sigma}{\rho l^2}}$$

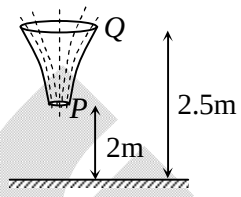
16. 1

Sol. $P_1 - P_2 = \frac{1}{2}\rho\left(v_2^2 - v_1^2\right) + \rho g\left(h_2 - h_1\right)$

$= \frac{1}{2}\times 10^3(1^2 - 3^2) + 10^3 \times 10(0.5)$

$= -4\times 10^3 + 5\times 10^3$

$= 1000\ \text{J/m}^3$



17. 4

Sol. $\lambda_A = \frac{0.69}{5}\text{day}^{-1}$ and $\lambda_B = \frac{0.69}{30}\text{day}^{-1}$

Given $\left(\frac{dN_{OA}}{dt}\right) = 64\left(\frac{dN_{OB}}{dl}\right)$

$\therefore \quad \lambda_A N_{OA} = 64\lambda_B N_{OB}$

At time t, $\frac{dN_A}{dt} = \frac{dN_B}{dt}$

$\therefore \quad \lambda_A N_A = \lambda_B N_B$

Dividing equation (1) and (2) $\frac{N_{OA}}{N_A} = 64\left(\frac{N_{OB}}{N_B}\right)$

But $N_A = N_{OA}\theta^{-2\lambda^1}$ and $N_B = N_{OB}\theta^{-2B^1}$

$\therefore \quad$ Equation (3) becomes $\theta^{\lambda}\lambda^1 = 64e^2e^t$

$\therefore \quad {}_a(\lambda_A - \lambda_B)t = 64$

$\therefore \quad (\lambda_A - \lambda_B)t = \ln(64) = 2.303\log_{10} 64$

$\therefore \quad t = \frac{2.303\log_{10} 64}{\left(\frac{0.693}{5} - \frac{0.693}{30}\right)} = 36\ \text{days}$

18. 6

Sol. $AP = 1 = s - a$

$AQ = 6 = s$

$\therefore s - (s - a) = 6 - 1 \Rightarrow a = 5$

$a = 2R\sin A$

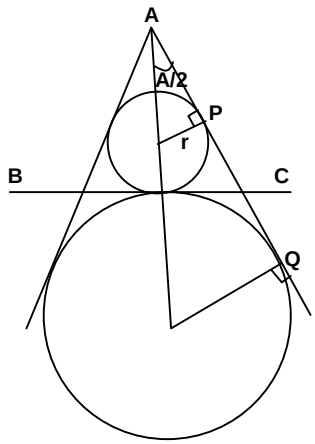
$\Rightarrow \sin A = 1$

$\Rightarrow A = 90^o$

$\therefore \tan(A/2) = \frac{r}{1}$

$\Rightarrow r = 1$

$\therefore \Delta = s.r = 6$



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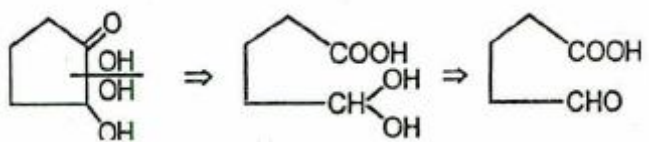
Chemistry

PART – II

Section – A

19. A

Sol.



20. A

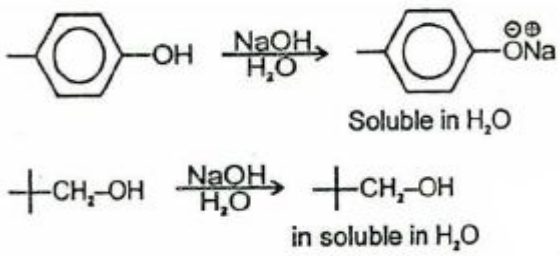
Sol. $Z = \frac{PV}{RT}$ $\left(\text{for 1 mole gas}\right)$

$$0.869 = \frac{71\text{bar} \times V}{\left(0.0821\text{L} - \text{atm R mol}^{-1}\right) \times \left(400\text{K}\right)}$$

$$V = \frac{0.869 \times 0.0821 \times 1.01325 \times 400}{71} = 0.407 \simeq 0.41\,\text{L}$$

21. C

Sol.



22. D

Sol. $H \rightarrow HF, Cl \rightarrow ClF, ClF_3, ClF_5$

$Br \rightarrow BrF, BrF_3, BrF_5$

$I \rightarrow IF, IF_3, IF_5, IF_7$

23. A, B, C, D

Sol. $\Rightarrow 2Ca(OH)_2 + Cl_2(g) \xrightarrow{40^\circ C} \underset{(x)}{CaCl_2} + Ca\left(\underset{(y)}{OCl}\right)_2 + 2H_2O$

$\Rightarrow CaCl_2 + Ca(OCl)_2 + \underset{\text{excess}}{H_2O} \longrightarrow Ca(OH)_2 + Cl_2$

$\Rightarrow$ Oxidizing properties of $Ca(OCl)_2$ is due to presence of OCl^-

$\Rightarrow$ 'X' and 'Y' exist together in the presence of basic medium.

24. A, C, D

Sol. $S = \frac{1}{2}$ and $-\frac{1}{2}$

Spin quantum number is not obtained by schrodinger wave equation orbital angular momentum $= \sqrt{\ell(\ell+1)}.\frac{h}{2\pi}$

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25. A, B

Sol. Monosaccharides are reducing sugars. Acetals (as D) are non reducing sugars.

26. A, C

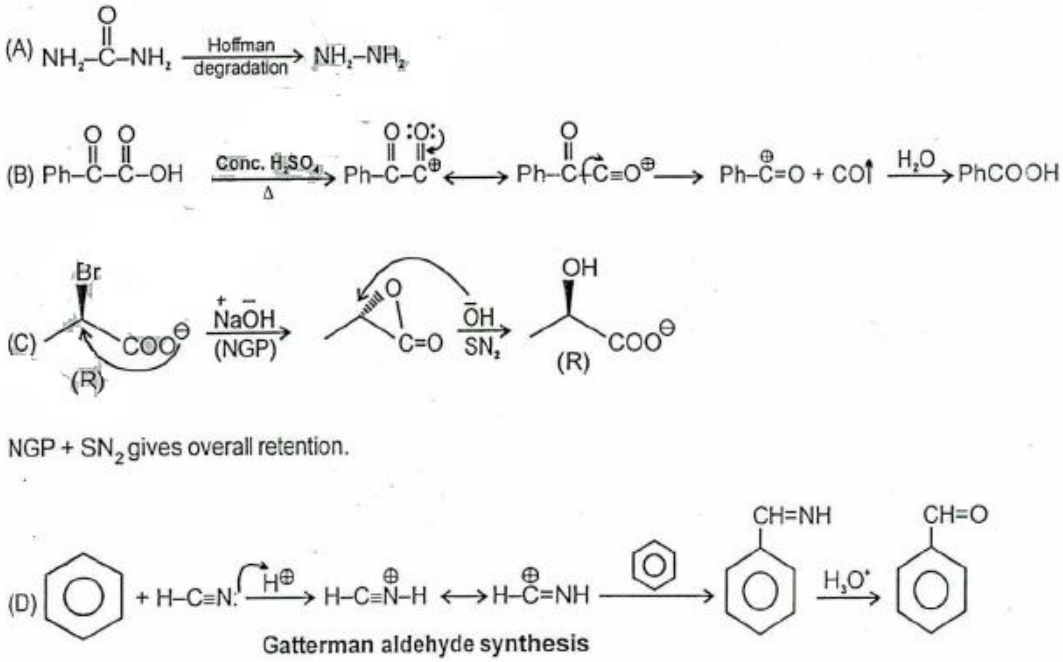
Sol. In NaCl structure, $\frac{r_+}{r_-} = 0.414$

In antifluorite structure S^{2-} ions present at alternate tetrahedral interstitial site of fcc. Hence coordination number of anion is 4.

In CsCl (BCC lattice), $2(r_+ + r_-), 2(r_+ + r_-) = \sqrt{3}a$

27. A, B, C

Sol.



28. A, B, D

Sol. $Mg + NH_3(g) \xrightarrow{900^\circ C} Mg_3N_2 + H_2(gas)$

$Mg_3N_2 + H_2O \longrightarrow Mg(OH)_2 \downarrow + NH_3(gas)$

Section – B

29. 4

Sol. $K_c = \frac{[z]^2}{[y][x]} = \frac{\left(\frac{3}{4}\right)^2}{[y]}; \frac{9}{16} = \frac{9}{16} \times \frac{1}{[y]}$

$\therefore [y] = 1$, moles of y = 4 at equilibrium

30. 6

Sol.

$2NaOH$	+	$NiCl_2$	$\longrightarrow$	$Ni(OH)_2$	+	$2NaCl$
2 mole		2 mole				
–		1 mole		1 mole		2 mole

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At equilibrium, $\left[Ni^{2+}\right] = 1M$

$K_{sp}\left[Ni(OH)_2\right] = 10^{-16} = \left[Ni^{2+}\right]\left[OH^-\right]^2, \left[OH^-\right]^2 = 10^{-16}$

$\therefore pOH = 8, pH = 6$

31. 7

Sol $Ag_2S, PbCrO_4, CaC_2O_4, FePO_4, CuS, PbI_2, Ag_2S_2O_3$

32. 1

Sol. $nC_{v,m}\left[T_2 - T_1\right] = -0.821\left[V_2 - V_1\right]$

$0.5 \times 2R\left[T_2 - T_1\right] = -0.821[7-2]$

$0.0821\left[T_2 - T_1\right] = -0.821 \times 5$

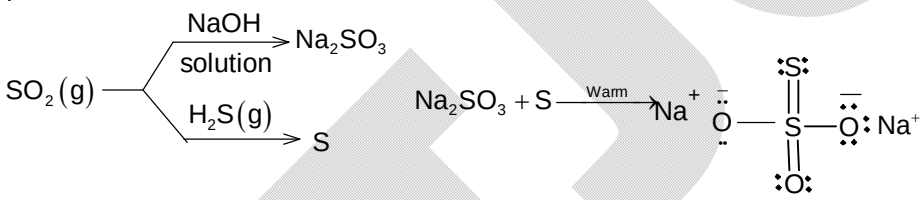
$T_2 - T_1 = -50$

$\Delta S_{total} = \Delta S_{system} + \Delta S_{surr} = nC_{v,m} \ln\left(\frac{T_2}{T_1}\right) + nR \ln\left(\frac{V_2}{V_1}\right) \quad (\Delta S_{surr} = 0)$

$= 0.5 \times 2 \ln\left(\frac{7}{2}\right) + 0.5 \times 2R \ln\left(\frac{350}{400}\right) = 1.95 - 0.7 \times (1.95 - 2.1) = 0.95 \ll 1$

33. 4

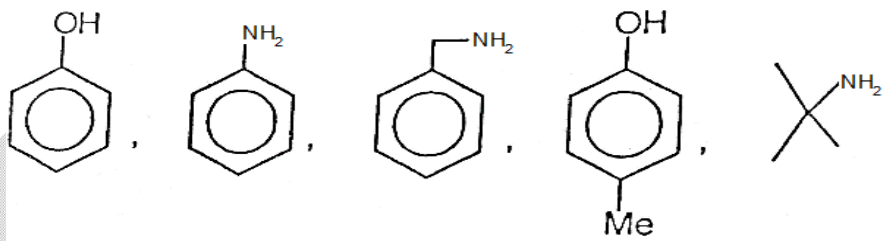
Sol.



Algebric sum of oxidation states of sulphur atoms present in a molecule of compound 'Z' = + 6 – 2= + 4

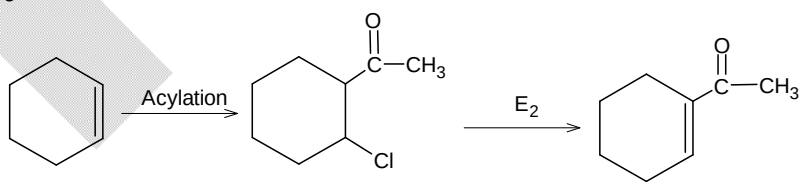
34. 5

Sol.



35. 0

Sol.



36. 7

Sol. $Pb_2O_3, ClO_2, P_4O_8, GaO, Fe_3O_4, Pb_3O_4, NO_2$

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Mathematics

PART – III

Section – A

37. C

Sol. $T_{r+1} = {}^{25}C_r (ab)^{h-1} (x)^{\frac{h-r}{2}} c^r (x)^{-r/3}$

$h = 25 \Rightarrow r = 15$

$\Rightarrow {}^{25}C_{15} a^{10} b^{10} c^{15}$

$10a + 10b + 15c = 1$

Use AM $\geq$ GM inequality

Required value = ${}^{25}C_{15} \left(\frac{1}{35}\right)^{35}$

38. B

Sol. Put $\sqrt{x} = \tan^2\theta \qquad \Rightarrow x = \tan^4\theta$

$dx = 4\tan^3\theta.\sec^2\theta d\theta$

$I = \int \tan\theta.4\tan^3\theta.\sec^2\theta.d\theta = \left(\frac{4}{5}\right)x^{5/4} + c$

39. C

Sol. Vector which is orthogonal to $\vec{a}$ and caplanar to $\vec{b}$ and $\vec{c}$ is $\vec{a}\times(\vec{b}\times\vec{c})$

$\therefore$ Required vector is $\frac{3\hat{j}-\hat{k}}{\sqrt{10}}$

40. A

Sol. Differential equation can be rewritten as

$$\left(\frac{xdy-ydx}{x^2}\right)\left(\frac{x^2}{y^2}\right) = (x\,dy + y\,dx)\sin xy$$

41. A, C, D

Sol. $z^2 + az + a^2 = 0$

$\Rightarrow$ $z = a\omega,\ a\omega^2$ (where 'ω' is non real root of cube unity)

$\Rightarrow$ locus of z is a pair of straight lines

and arg (z) =arg(a) + arg(ω) or arg(a) + arg(ω^2)

$\Rightarrow \arg(z) = \pm\frac{2\pi}{3}$

also, $|z| = |a||\omega|$ or $|a|\,|\omega^2| \Rightarrow |z| = |a|$.

42. A, C

Sol. Let $a_1 = (2m+1)^2$, $a_2 = (2n+1)^2$

$\Rightarrow a_1 - a_2 = 4(m(m+1)-n(n+1)) = 8k$

so, difference of any two odd square is always a multiple of 8

Now apply $C_1 - C_3$ and $C_2 - C_3$

then C_1 and C_2 both become multiple of 8 so Δ always a multiple of 64.

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43. A, D

Sol. Since $\int_a^b f(x)dx = (b-a)\int_0^1 f\{(b-a)x+a\}dx$,

$$\int_1^2 \sin x^2 dx = \int_0^1 \sin(x+1)^2 dx = \int_0^1 \sin(x^2+2x+1)dx$$

$$\int_{-4}^{4} \cos x^2\, dx = 8\int_0^1 \cos(8x-4)^2\, dx$$

$$= 8\int_1^0 \cos 16(2x-1)^2\, dx.$$

44. C, D

Sol. For a > b > c > d > e > f

Required cases $= {}^{10}C_6$ or ${}^{10}C_4$

For a < b < c < d < e < f, required cases ${}^{9}C_6 = {}^{9}C_3$

For a < b ≤ c < d < e < f, we here a < b < c < d < e < f or a < b = c < d < e < f

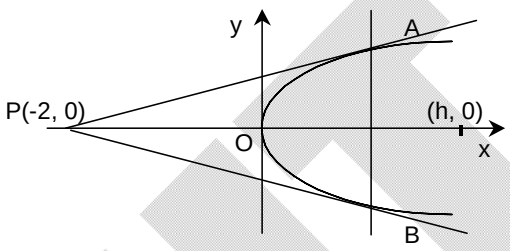
Total cases = ${}^{9}C_6 + {}^{9}C_5 = {}^{10}C_6$ or ${}^{10}C_4$

for a < b ≤ c < d ≤ e < f, we have a < b < c < d < e < f or a < b = c < d < e < f
or a < b < c < d = e < f or a < b = c < d = e < f

${}^{9}C_6 + {}^{9}C_5 + {}^{9}C_5 + {}^{9}C_4 = {}^{10}C_6 + {}^{10}C_5 = {}^{11}C_6$ or ${}^{11}C_5$

45. A, B

Sol.



Point 'P' clearly lies on the directrix of $y^2 = 8x$.
Thus slope of PA and PB are 1 and -1 respectively.
Equation of PA : y =x +2, equation of PB : y = -x –2, equation of AB : x = 2.
Let the centre of the circle be (h, 0) and radius be 'r'

$$\Rightarrow \frac{|h+2|}{\sqrt{2}} = \frac{|h-2|}{1} = r$$

$\Rightarrow$ $h^2 + 4 + 4h = 2(h^2 + 4 - 4h) \Rightarrow h^2 - 12h + 4 = 0$

$h = \frac{12 \pm 8\sqrt{2}}{2} = 6 \pm 4\sqrt{2} \Rightarrow |h-2| = 4(\sqrt{2}-1),\ 4(\sqrt{2}+1)$.

46. A, B, C, D

Sol. $(\text{adjB})\,\text{adj}(\text{adjB}) = |\text{adj B}|\text{I}$

$(\text{B adjB})\,\text{adj}(\text{adjB}) = |\text{adj B}|\text{B}$

$\Rightarrow |\text{B}|\,\text{adj}(\text{adj B}) = |\text{B}|^{n-1}\text{B}$

$\Rightarrow \quad \text{adj}(\text{adjB}) = |\text{B}|^{n-2}\text{B}$

$(\text{adj A})^{\text{T}} = (|\text{A}|\text{A}^{-1})|^{\text{T}} = |\text{A}|(\text{A}^{\text{T}})^{-1} = |\text{A}|\text{A}^{-1} = \text{adj A}$

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Section – B

47. 4

Sol. $\lim\limits_{x\to0}\dfrac{\int_0^x t^2dt}{(x-\sin x)\sqrt{a+t}} = \lim\limits_{x\to0}\dfrac{\int_0^x \frac{t^2dt}{\sqrt{a+t}}}{(x-\sin x)} = \lim\limits_{x\to0}\dfrac{x^2}{\sqrt{a+x}\,(1-\cos x)}$

$= \lim\limits_{x\to0}\dfrac{x^2}{2\sin^2\frac{x}{2}.\sqrt{a+x}} = \dfrac{2}{\sqrt{a}}$ =1 (given)

$\Rightarrow$ a = 4.

48. 6

Sol. $S - S_n < \dfrac{1}{300}$

$\Rightarrow \dfrac{1}{1-\frac{1}{3}} - \dfrac{1-\frac{1}{3^n}}{1-\frac{1}{3}} < \dfrac{1}{300} \Rightarrow \dfrac{3}{2}\left[1-1+\dfrac{1}{3^n}\right] < \dfrac{1}{300}$

$\Rightarrow \dfrac{1}{3^n} < \dfrac{1}{300}\times\dfrac{2}{3} \Rightarrow \dfrac{1}{3^n} < \dfrac{1}{450} \Rightarrow 3^n > 450$

$\therefore$ least value of n = 6

49. 8

Sol. $\cos ec^2\alpha = \cos ec^2\beta = \cos ec^2\gamma = 1$

α,β,γ many be $\dfrac{\pi}{2}, \dfrac{3\pi}{2}$

Hence no. of ordered pairs = $2\times2\times2$ =8

50. 4

Sol. Let sides of rectangle be p and q; Area of rectangle $= pq = 200$ …….(i)

Area of ellipse $= \pi ab = 200\pi \therefore ab = 200$ ……(ii)

Distance BD $= \sqrt{p^2+q^2}$ = distance between foci $\Rightarrow p^2+q^2 = 4a^2e^2$ or

$(p+q)^2 - 2pq = 4(a^2-b^2)$ ……(iii)

Also from the definition of ellipse sum of focal lengths is 2a. Then AB + AD = p + q = 2a (iv)

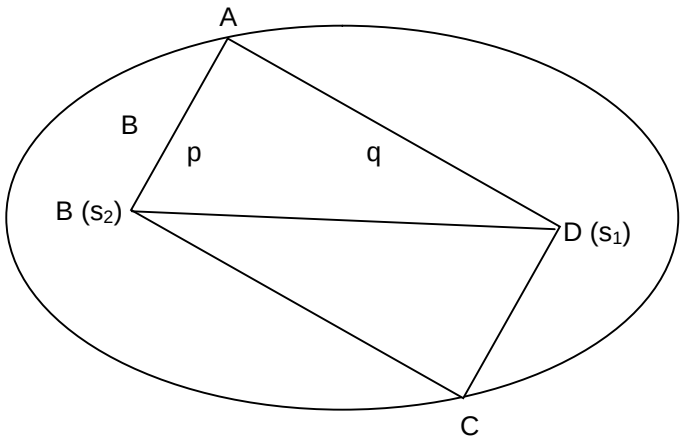
putting value of $(p+q)$ in equation (3) from equation (4) we have $(2a)^2 - 2pq = 4a^2 - 4b^2$

$\Rightarrow 4a^2 - 2\times200 = 4(a^2-b^2)$ $(\because pq = 200) \Rightarrow a^2 - 100 = a^2 - b^2 \Rightarrow b = 10$ from equation (2),

$ab = 200 \Rightarrow a = 20$ since, $p+q = 2a$ (from (4))

$\therefore$ perimeter = $= 2(p+q) = 4a = 4\times20 = 80$

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51. 6

Sol. $f(x)=(x-a)^3+b$

$\because$ Rolle's theorem is applicable to $\phi(x)$ at $x=2$

$\therefore\ \phi(a)=\phi(b)$ and $f'(2)=0$

$\Rightarrow\ 0=f(b)-f(a)+(a-b)f'(b)+3(b-a)^2$

And $(a-2)f''(2)+6(2-a)=0$

$(a-2)f''(2)+6(2-a)=0$ and $f''(2)=6$

52. 8

Sol. $\lim\limits_{x\to0} I_1=\dfrac{\lim\limits_{x\to0}\int_0^x \dfrac{t^2dt}{\sqrt{\lambda^2-2\lambda t}}}{(x-\sin x)}=\lim\limits_{x\to0}\dfrac{x^2}{(1-\cos x)\sqrt{\lambda^2+2\lambda x}}=1$

$\Rightarrow \dfrac{2}{|\lambda|}=1 \qquad \Rightarrow |\lambda|=2$

$\Rightarrow \lambda=\pm2$

53. 5

Sol. $\tan\dfrac{\pi}{15}+\tan\dfrac{4\pi}{15}+\tan\dfrac{7\pi}{15}+\tan\dfrac{10\pi}{15}+\tan\dfrac{13\pi}{15}$

$=(\tan12^o+\tan48^o)+(\tan84^o-\tan24^o)-\sqrt{3}$

$=\dfrac{\sin60^o}{\cos12^o\cos48^o}+\dfrac{\sin60^o}{\cos24^o\cos84^o}-\sqrt{3}$

$=\sqrt{3}\left[\dfrac{1}{\cos60^o+\cos36^o}+\dfrac{1}{\cos108^o+\cos60^o}-1\right]$

$=\sqrt{3}\left[\dfrac{1}{\frac{1}{2}+\frac{\sqrt{5}+1}{4}}+\dfrac{1}{\frac{1}{2}-\frac{\sqrt{5}-1}{4}}-1\right]=5\sqrt{3}$

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54. 4

Sol. Let a be the edge of the cube, so that

$\overrightarrow{OA} = a\hat{i}$, $\overrightarrow{OB} = a\hat{j}$ and

$\overrightarrow{OC} = a\hat{k}$; $\overrightarrow{OL} = \overrightarrow{OB} + \overrightarrow{BL} = \overrightarrow{OB} + \overrightarrow{OA} = a\hat{i} + a\hat{j} = a\left(\hat{i} + \hat{j}\right)$

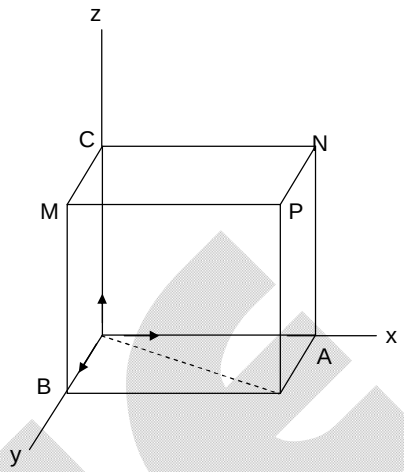
Similarly, we can have $\overrightarrow{OM} = a\left(\hat{j} + \hat{k}\right)$; $\overrightarrow{ON} = a\left(\hat{k} + \hat{i}\right)$;

and $\overrightarrow{OP} = a\left(\hat{i} + \hat{j} + \hat{k}\right)$

Now $\overrightarrow{CL} = \overrightarrow{OL} - \overrightarrow{OC} = a\left(\hat{i} + \hat{j}\right) - a\hat{k} = a\left(\hat{i} + \hat{j} - \hat{k}\right)$;

$\overrightarrow{AM} = \overrightarrow{OM} - \overrightarrow{OA} = a\left(\hat{j} + \hat{k}\right) - a\hat{i} = a\left(\hat{j} + \hat{k} - \hat{i}\right)$

and $\overrightarrow{BN} = \overrightarrow{ON} - \overrightarrow{OB} = a\left(\hat{k} + \hat{i}\right) - a\left(\hat{k} + \hat{i} - \hat{j}\right)$



Let Q be a point $x\hat{i} + y\hat{j} + z\hat{k}$ on a line drawn through O parallel to given line, so that

$\overrightarrow{OQ} = x\hat{i} + y\hat{j} + z\hat{k} \therefore OQ = \sqrt{x^2 + y^2 + z^2}$

Now $\overrightarrow{OP}.\overrightarrow{OQ} = OP.OQ\cos\alpha$

$\Rightarrow \left(\hat{i} + \hat{j} + \hat{k}\right).a\left(x\hat{i} + y\hat{j} + z\hat{k}\right) = a\sqrt{1+1+1}\sqrt{x^2 + y^2 + z^2}\cos\alpha$

$\Rightarrow a\left(x + y + z\right) = a\sqrt{3}\ \sqrt{x^2 + y^2 + z^2}\cos\alpha \quad \Rightarrow \quad \cos\alpha = \dfrac{x + y + z}{\sqrt{3}\sqrt{x^2 + y^2 + z^2}}$

Similarly, $\cos\beta = \dfrac{x + y - z}{\sqrt{3}\sqrt{x^2 + y^2 + z^2}}$; $\cos\gamma = \dfrac{-x + y + z}{\sqrt{3}\sqrt{x^2 + y^2 + z^2}}$ and $\cos\delta = \dfrac{x - y + z}{\sqrt{3}\sqrt{x^2 + y^2 + z^2}}$

Now, $\cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta$

$$\frac{\left(x + y + z\right)^2 + \left(x + y - z\right)^2 + \left(-x + y + z\right)^2 + \left(x - y + z\right)^2}{3\left(x^2 + y^2 + z^2\right)} = \frac{4\left(x^2 + y^2 + z^2\right)}{3\left(x^2 + y^2 + z^2\right)} = \frac{4}{3} \Rightarrow \lambda = 4$$

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